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| <b>2.5 Practical investigation</b> | <p>2.5.1 One practical research exercise to gather data relevant to topics covered in cognitive psychology. This practical research exercise must adhere to ethical principles in both content and intention.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                    | <p>In conducting the practical research exercise, students must:</p> <ul style="list-style-type: none"> <li>• design and conduct a laboratory experiment to gather quantitative data and include descriptive statistics as analysis and a non-parametric test of difference</li> <li>• make design decisions when planning and conducting your experiment, including experimental design, sampling decisions, operationalisation, control, ethical considerations, hypothesis construction, experimenter effects and demand characteristics</li> <li>• collect, present and comment on data gathered, including using measures of central tendency (mean, median, mode as appropriate); measures of dispersion (including range and standard deviation as appropriate); bar graph, histogram, frequency graph as relevant; normal distribution if appropriate and draw conclusions</li> <li>• use a Mann-Whitney U or Wilcoxon non-parametric test of difference to test significance (as appropriate), including level of significance and critical/observed values</li> <li>• consider strengths and weaknesses of the experiment, and possible improvements</li> <li>• write up the procedure, results and discussion section of a report.</li> </ul> |
|                                    | <p>Suitable examples</p> <ul style="list-style-type: none"> <li>• Dual task experiment to investigate components of working memory.</li> <li>• An experiment to look at acoustic similarity of words and the effect on short-term memory.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |